

NULL Values & 3-Valued Logic

Q. Explain the concept of NULL in SQL. How do we check for NULL values, handle them in expressions, and how do they affect comparisons and aggregate functions?

****Definition to Remember****

****NULL****
an empty
with NU

1. Key Rules About NULL

* ****NULL != 0**** (zero is a number; NULL is the absence of a value)

* ****NU**

2. Checking for NULL Values

You cannot use ``= NULL`` or ``!= NULL`` - these always return UNKNOWN. Use specialized operators:

sql

-- IS NU

-- IS NOT NULL: find rows where a column has a value

SELEC

3. Three-Valued Logic

SQL uses True, False, and Unknown (instead of just True/False):

| Condition | Result |

| :--- | :-

4. Handling NULL with COALESCE()

`COALESCE(val1, val2, ...)` returns the ****first non-NULL value**** in the list - used to substitute a default value for NULLs.

sql

5. Effect on Aggregate Functions

* ``SUM()`, `AVG()`, `MAX()`, `MIN()`` -> ****automatically ignore NULL rows****.

* ``CO`

Example: Salaries = [100, 200, NULL] -> ``AVG(Salary)`` = (100+200)/2 = 150 ***(NULL ignored)***

*****Must-Write Points (for 10 marks)*****

1. NULL

*****Quick Recall*****

`NULL` =
COALE